

2081

Bachelor of Science in Computer Science and Information Technology

Course Title: **Mathematics I**

Code No: **MTH117** Semester: **1st Semester**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Section A

1. (a) Sketch the graph of $f(x) = x^2$. Find its domain and range. (b) Evaluate: $\lim_{x \rightarrow 1} \sin^{-1} \left(\frac{1 - \sqrt{x}}{1 - x} \right)$
2. (a) Where the function $f(x) = |x|$ is differentiable? Discuss. (b) A farmer has 1200 m. of fencing and wants to fence off a rectangular field that borders a straight river. He needs to fence along the river. What are the dimensions of the field that has the largest area?
3. (a) Find the solution of the initial value problem $(x^2 y' + xy = 1)$, $(y(1) = 2)$, $(x > 0)$. (b) Find the area enclosed by the line $(y = x - 1)$ and the parabola $(y^2 = 2x + 6)$.

Section B

Attempt any Eight questions [8×5=40]

4. Evaluate: $\int_0^{\sqrt{3}} \sqrt{1 + x^2} \, dx$.
5. Find the Maclaurin series expansion of $f(x) = \sin x$ for all x .
6. Find the unit normal and binormal vectors for the circular helix $\mathbf{r}(t) = \cos t \, \mathbf{i} + \sin t \, \mathbf{j} + t \, \mathbf{k}$.
7. If $f(x, y) = \frac{xy}{x^2 + y^2}$, does $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ exist? Justify.
8. Determine whether the sequence $(a_n = (-1)^n)$ is convergent or divergent.
9. The position vector of an object moving in a plane is given by $\mathbf{r}(t) = t^3 \mathbf{i} + t^2 \mathbf{j}$. Find its velocity, speed, and acceleration when $(t = 1)$ and illustrate geometrically.
10. Show that every member of the family of function $(y = \frac{1 + ce^t}{1 - ce^t})$ is a solution of the differential equation $(y' = \frac{1}{2}(y^2 - 1))$.
11. If $f(x, y) = 2x^3 - x^2y^3 - y^4$, find $(f_x(1, -2))$, $(f_y(1, -1))$ and $(f_{yx}(1, -1))$.
12. Use cylindrical shells to find the volume of the solid obtained by rotating about the x-axis the region under the curve $(y = \sqrt{x})$ for 0 to 1.